



UNIVERSITY OF GHANA

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DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

ACTU 304: LIFE CONTINGENCIES II (3 credits)

EXERCISES ON LIFE ANNUITIES

1. You are given:

- i. $T(x)$ is the random variable for the future lifetime of (x) .
- ii. $\mu(x+t) = \mu$.
- iii. $\delta = \mu$.

Determine $Var(\bar{a}_{\overline{T_x}|})$ in terms of μ .

2. For a continuous whole life annuity of 1 on (x) :

- i. $T(x)$ is the future lifetime random variable for (x) .
- ii. The force of interest and force of mortality are equal and constant.
- iii. $\bar{a}_x = 12.5$.

Calculate the standard deviation of $\bar{a}_{\overline{T_x}|}$.

3. You are given:

- i. $\mu = \delta = c$, where c is a positive constant.
- ii. ${}^2\bar{a}_x = \frac{25}{3}$

Calculate $Var(\bar{a}_{\overline{T_x}|})$

4. You are given:

i. $\delta = 0.05$.

ii. $\mu_x = \begin{cases} 0.05 & 0 \leq x < 50 \\ 0.08 & x \geq 50 \end{cases}$.

Calculate the actuarial present value of a continuous whole life annuity of 1 per year on (30).

5. Assume: $\mu_x(t) = 0.02$ for $t > 0$ and $\delta = 0.05$. Calculate $\ddot{a}_{x:\overline{15}|}$.

6. You are given:

i. ${}_k|q_{35} = 0.005(k+1)$ for $k = 0, 1, 2, 3$.

ii. $i = 0.05$.

Calculate the actuarial present value of a 4-year temporary life annuity-due of 100 per year on (35).

7. You are given the following probability mass function of $K(x)$:

k	0	1	2
$Pr(K_x = k)$	0.2	0.3	0.5

Find the variance of the present value random variable for a whole life annuity due if $i = 0.05$.

8. You are given:

x	q_x	$\ddot{a}_x$
75	0.03814	7.4927
76	0.04196	7.2226

Calculate the interest rate i .